

MISSOURI

S&T

MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 1.8

Linear Transformations

Transformations

A transformation (or function) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a corresponding vector $T(\mathbf{x})$ in $\mathbb{R}^m$.

$\mathbb{R}^n$ is the domain of T .

$\mathbb{R}^m$ is the codomain of T .

$T(\mathbf{x})$ is called the image of $\mathbf{x}$ under the action of T .

The range of T is the set of all possible images $T(\mathbf{x})$.

Example 1

Identify the domain, codomain, and range of the transformation $f(x) = x^2$.

Matrix Transformations

A matrix transformation is a transformation defined as $T(\mathbf{x}) = A\mathbf{x}$ where A is an $m \times n$ matrix.

$\mathbb{R}^n$ is the domain of T .

$\mathbb{R}^m$ is the codomain of T .

The range of T is the set of all linear combinations of the columns of A .

Example 2

Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$.

Find the images under T of

$\mathbf{u} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$.

Example 3

Find a vector $\mathbf{x}$ whose image under $T(\mathbf{x}) = A\mathbf{x}$ is $\mathbf{b}$, and determine whether $\mathbf{x}$ is unique.

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 3 & -8 & 8 \\ 0 & 1 & 2 \\ 1 & 0 & 8 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 6 \\ 3 \\ 10 \end{bmatrix}$$

Linear Transformations

A transformation T is linear if

- i. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T , and
- ii. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all vectors $\mathbf{u}$ in the domain of T .

Every matrix transformation is a linear transformation, but not all linear transformations are matrix transformations.

We're Doing The Same Thing Again!

Old:

Solve $A\mathbf{x} = \mathbf{0}$.

New:

Find all $\mathbf{x}$ in $\mathbb{R}^n$ that are mapped into the zero vector by the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

We're Doing The Same Thing Again!

Old:

Solve $A\mathbf{x} = \mathbf{b}$.

New:

Is $\mathbf{b}$ in the range of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$?

Example 4

Plot $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ and their images under the transformation $T(\mathbf{x}) = A\mathbf{x}$ for $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
Describe the transformation geometrically.

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is called the 2×2 identity matrix I_2 .

Example 5

Plot $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ and their images under the transformation $T(\mathbf{x}) = A\mathbf{x}$ for $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
Describe the transformation geometrically.

Example 6

Plot $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ and their images under the transformation $T(\mathbf{x}) = A\mathbf{x}$ for $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$.
Describe the transformation geometrically.

Properties of Linear Transformations

If T is a linear transformation, then

$$T(\mathbf{0}) = \mathbf{0}$$

and

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$

for all vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars c, d .

Example 7

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Find the images of $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
